

- Q.1a) a, b, c are three events in a distributed system and no two events belong to the same process. Using Lamport's definition of sequential and concurrent events, comment on the truth of the following statements:
- (i) $(a \parallel b) \wedge (b < c) \Rightarrow a < c$
 - (ii) $(a \parallel b) \wedge (b \parallel c) \Rightarrow a \parallel c$ (Notation: $a \parallel b$ denotes that a, b are concurrent events.)
 - (iii) Which set of events is concurrent ? **A**) [(3, 1, 5, 7), (3, 2, 6, 7), (2, 1, 6, 8)], **B**) [(2, 1, 3, 4), (2, 2, 3, 3), (3, 3, 2, 5)], **C**) [(1, 2, 3, 4), (2, 3, 4, 5), (3, 4, 5, 6)], **D**) (1, 5, 6, 7), (1, 4, 5, 7), (1, 3, 2, 2).
- b) Since the size of vector clock grows linearly with the number of processes N, is it possible to detect causality or concurrency using clocks of size smaller than N? Justify your answer.
- c) What are version vectors, and how do they ensure eventual consistency in distributed systems?
- d) How does vector clocks can enforce causal communication? **[CO1: 3+2+3+2=10]**
- Q.2a) In Ricart–Agrawala algorithm the critical section is accessed in increasing order of timestamp. Does the same hold in Maekawa's algorithm? .
- b) Give a situation where Maekawa's algorithm can enter into a deadlock.
- c) What is the response time of the Token based Mutual Exclusion algorithm under low load and high load conditions?
- d) What is a consistent cut in the context of distributed systems, and why is it important for ensuring system consistency? **[CO3: 2+3+2+3=10]**
- Q.3a) In what respect are distributed computing better than parallel processing system?
- b) Define Uniform, Non-uniform, Anonymous, Non-Anonymous, oriented and non-oriented Rings.
- c) Illustrate the best case and worst case scenarios in the Ring Election Protocol and also determine the message complexity in each case.
- d) How does Zookeeper handle failures during the Leader election process? **[CO2 : 2+3+3+2 = 10]**